

Recitation 10

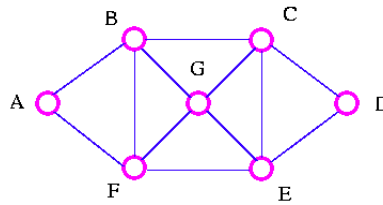
Graph Theory

Definitions

- Let $G = (V, E)$ and $G' = (V', E')$. G and G' are **isomorphic** if there is a bijection $f : V \rightarrow V'$ such that for every pair of vertices $\{x, y\} \in E$ if and only if $\{f(x), f(y)\} \in E'$. In other words, two graphs are isomorphic if we can find a consistent relabeling of the edges and vertices of the two graphs.
- A **walk** from v_0 to v_1 in an undirected graph G is a sequence of alternating vertices and edges that starts at v_0 and ends with v_1 .
- An **open walk** is a walk in which the first and last vertices are not the same.
- A **closed walk** is a walk in which the first and last vertices are the same.
- A **trail** is an open walk in which no edge occurs more than once.
- A **circuit** is a closed walk in which no edge occurs more than once.
- A **path** is a trail in which no vertex occurs more than once.
- A **cycle** is a circuit in which no vertex occurs more than once, except the first and last vertices which are the same.
- Two vertices are **connected** if there exists a path between them (or if they are the same vertex).
- An **Euler trail** is a trail that visits every edge exactly once.
- An **Euler circuit** is a circuit that contains every edge exactly once.
- A **Hamiltonion circuit** in an undirected graph is a circuit that includes every vertex (except first/last vertex) exactly once.
- A **Hamiltonion path** in an undirected graph is a path that includes every vertex in the graph.
- A **Bipartite graph** is a graph whose vertex set can be partitioned to two parts, and no edge connects a pair of vertices belonging to the same part. Any graph that is a subgraph of a complete bipartite graph is bipartite.

Task 1

Consider the following road map:



- a. An explorer wants to explore all the routes between a number of cities. Can a tour be found which traverses each route only once? Particularly, find a path which starts at A, goes along each road exactly once, and ends back at A.

In other words, find an Euler trail for the graph!

Many examples, one possibility is $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow B \rightarrow G \rightarrow C \rightarrow E \rightarrow G \rightarrow F \rightarrow A$. The Explorer travels along each road (edges) just once but may visit a particular city (vertex) several times.

- b. A traveler wants to visit a number of cities. Can a path be found which visits each city only once? Particularly, find a tour which starts at A, goes to each city exactly once, and ends back at A.

In other words, find a Hamiltonian cycle for the graph!

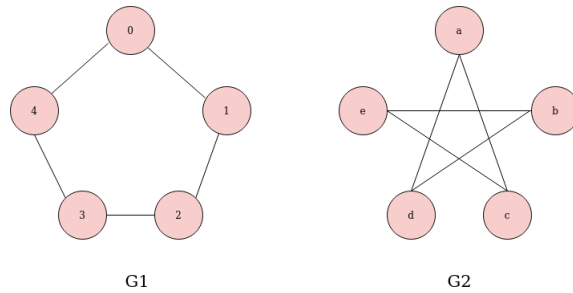
One such solution is $A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow G \rightarrow F \rightarrow A$. The travelers visits each city (vertex) just once but may omit several of the roads (edges) on the way.

- c. *Optional:* Remember the n-dimensional cube graph (See the lectures or Task 7). Find a Hamiltonian cycle in Q_3 . Does Q_3 have an Eulerian circuit?

One such solution is $000 \rightarrow 001 \rightarrow 011 \rightarrow 010 \rightarrow 110 \rightarrow 111 \rightarrow 101 \rightarrow 100 \rightarrow 000$ (This sequence is also known as a **Gray code**). There is no Eulerian circuit because degrees are all odd.

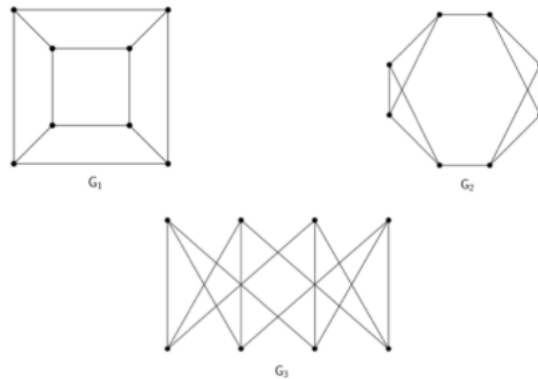
Task 2

Two graphs are isomorphic if there exists a matching or relabelling of the vertices of the graphs.



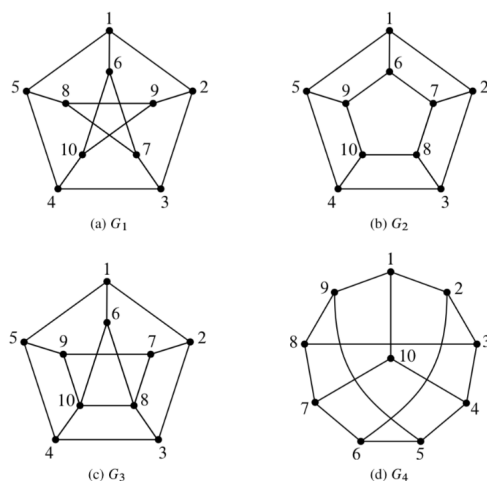
The two graphs above are isomorphic because we can match a with 0 , b with 3 , c with 1 , d with 4 , and e with 2 . Given this matching of the vertices, it's easy to check that the vertices are connected the same way in the graphs. Yet, it's hard to determine if two graphs are actually isomorphic! Let's try this ourselves:

1. Out of the three graphs below, determine which two are isomorphic.
Hint: Which of these are bipartite graphs?



G_1 and G_3 are isomorphic. An easy way to tell is that G_2 is the only graph with cycles of length 3. Similarly, G_1 and G_3 are both bipartite and thus can be colored using two colors, while G_2 cannot. This example can also be found on Wikipedia's [Graph Isomorphism page](#).

2. *Optional:* Let's try an even harder example! Determine which two out of these four graphs are isomorphic.



G_1 and G_4 are isomorphic, and they're known as Petersen's graph! G_2 and G_3 are not isomorphic with G_1/G_4 because they have cycles of length 4. Moreover, G_3 is the only graph with cycles of length 3, so it is not isomorphic with any other graphs.

Task 3

- a. Given a certain labelling of the vertices, how many distinct graphs can be made with 4 vertices?

For each pair of vertices, there exists a possible edge. Thus, there are $\binom{n}{2}$ possible edges. Each can be either in or not in the edge set of a graph, so there are $2^{\binom{n}{2}}$ possible graphs. When $n = 4$, this is equal to $2^{\binom{4}{2}} = 2^6 = 64$.

- b. How many graphs with 4 vertices exist if we don't count repetitions under isomorphism?

11, as counted in lecture 4/5. So, from the previously counted 64 graphs most of them are isomorphic.

- c. For any graph G , explain why

$$\sum_{v \in V(G)} \deg(v) = 2|E(G)|$$

The degree of a vertex is a count of the number of vertices it is adjacent to; hence, the number of edges which contain the vertex. This means that when we add up all the degrees, we are counting each edge twice since each edge is a set of two vertices.

- d. The Cafeteria is having an event where two customers that purchase a sandwich to split receive 10% off their order. The 35 CS22 crewmates are planning on taking advantage of this offer by having each crewmate split sandwiches with 3 other crewmates. Prove that this is impossible.

Let each crewmate be a vertex such that there exists an edge between two crewmates if and only if they are splitting a sandwich. Each of the 35 vertices has degree 3, so the sum of the degrees is 105. We just proved that this equals 2 times the number of edges, which means there are 52.5 edges. This isn't an integer, which is impossible.

- e. *Optional:* Prove that in any graph there exist two vertices with the same degree.

Hint: What are the possible degrees of a vertex in a graph with n nodes?

A vertex can have degree from 0 to $n - 1$. However, if a vertex has degree $n - 1$, there can't be a vertex with degree 0 because that vertex is adjacent to every other. So, in this case, there are $n - 1$ possible values for the degree of a vertex, and n vertices: by the pigeonhole principle, two must have the same degree. On the other hand, if there is no vertex of degree $n - 1$, then vertices can have degree 0 through $n - 2$: still $n - 1$ options, so by the pigeonhole principle two have the same degree.

Checkpoint 1: Call over a TA.

Trees!

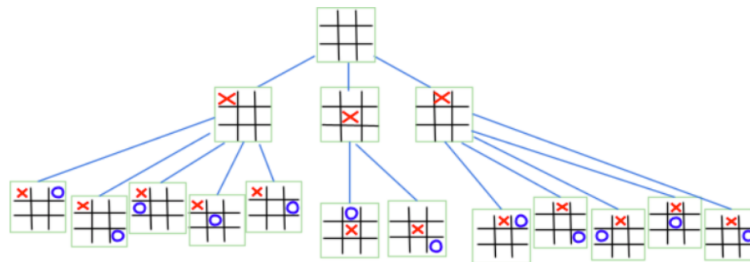
- A **cycle** is a path that starts and ends with the same vertex.
- A graph is **cyclic** if it contains a simple cycle and **acyclic** otherwise.
- A graph is **connected** if there is a simple path between each pair of vertices (that is, all vertices are connected).
- A **tree** is a connected, acyclic graph.
- A **forest** is an acyclic graph. In other words, a forest is a set of trees.
- A **leaf** of a tree is a vertex with degree 1.
- A **spanning tree** of a graph G is a subgraph T of G such that $V(T) = V(G)$ and T is a tree.

Task 4

In 1997, a chess-playing computer program called Deep Blue beat the reigning world champion of chess, Garry Kasparov, in a 6-round series. Although it was unclear whether computers can reason as well as humans, the victory did establish that computers are better at evaluating complex game trees, the basis of computer-based game strategies.

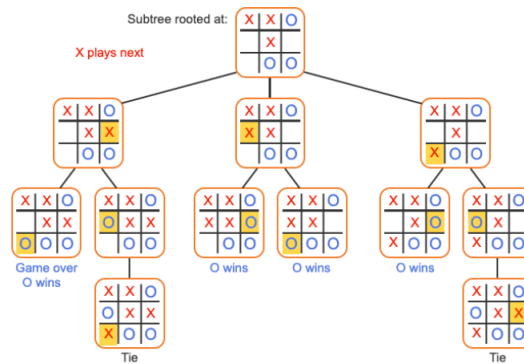
A game tree shows all possible playing strategies of both players in a game. Each vertex in the tree represents a configuration of the game as well as an indication of whose turn it is to play next. We illustrate game trees in a much simpler context of tic-tac-toe.

The first three layers of the game tree of tic-tac-toe



The **root** of the tree is the initial configuration of the game, which is an empty grid in the case of tic-tac-toe. The **children** of a configuration c are all the configurations that can be reached from c by a single move of the correct player. A configuration is a **leaf** vertex in the tree if the game is over.

Here, we see a subtree near the end of the game:



1. How many leaf nodes are in the subtree drawn above?

6

2. What is the length of the longest path from a root to a leaf in a tree represented by the full tic-tac-toe game?

9, since the longest tic-tac-toe game lasts 9 moves.

3. *Optional:* What is the length of the longest path between any two vertices in a tree represented by the full tic-tac-toe game?

18. The path would be from a leaf to another leaf after passing through the root node, which is $9 + 9 = 18$.

Task 5

1. Prove that in a tree, there is exactly one simple path between any two vertices.

The graph is connected, so there must be at least one path between any two vertices. Assume there were two paths between two vertices: there is a point where they stop being the same path, and a point where they rejoin at the end. If we put together the parts between these points, we get a cycle.

2. One way of defining a tree is as a **minimally connected, maximally acyclic graph**. This means that if we remove an edge, our graph will no longer be connected and if we add an edge, the graph will no longer be acyclic. Prove these two statements.

There is only one path between two vertices, so if we remove an edge $\{a, b\}$, there is now no way to get from a to b . All vertices are connected, so if we add the edge $\{a, b\}$, there are now two ways to get from a to b and hence a cycle.

3. *Optional:* Let T be a tree with at least two vertices. Prove that T has at least two leaves. *Hint:* Start by assuming that the longest simple path in T does not start and end with leaves. What contradiction can we reach?

Assume the longest simple path in T has a non-leaf endpoint. As all non-leaf vertices have degree greater than 1 and our path is simple (so the endpoint only appears once, there is at least one edge incident to the ending vertex which is unused. Extend the path by also traversing this edge. The path will still be simple because if we already used the new vertex, we would now have two ways of getting from the endpoint to the vertex. So, we have a longer simple path, which contradicts our assumption that our original path was longest. So, both end vertices are leaves.

Induction on Graphs

Many graph theory questions involve induction. When doing induction on graphs, we often use a technique referred to in CS22 as "build-down" induction. This is just regular induction, with a slightly different perspective. To review, here are the steps of an inductive proof:

1. Define the predicate $P(n)$.
2. Show that the base case is true.
3. Assume the inductive hypothesis is true. If you are using standard induction then you will assume $P(k)$ is true for some integer k . If you are using strong induction then you will assume $P(i)$ is true for all $i \leq k$.
4. Show that $P(k + 1)$ is true given the inductive hypothesis.

Here is where build down induction differs from our usual perspective. A lot of the time, we start by invoking our inductive hypothesis, and manipulating the k case to get the $k + 1$ case. In build down induction, we do the following instead:

- i. Consider an arbitrary $k + 1$ case.
- ii. Alter the $k + 1$ case to obtain an k case.

- iii. Invoke the induction hypothesis, demonstrating that your property holds for the k case.
- iv. Recover the original $k + 1$ case by undoing your alteration.
- v. Prove that the property still holds despite building back up to the original $k + 1$ case.

This technique is especially useful when it is not clear how to address all $k + 1$ cases in general when starting from an k case. For example, if we started with a graph with k edges, where do we add an edge to obtain a general graph with $k + 1$ edges? There are a lot of ways to add this extra edge that we have to consider. To avoid this, we can start with an arbitrary $k + 1$ case. Then, we build-down to an k case, at which point we can invoke our induction hypothesis. Then, we build back up to the *original* $k + 1$ case.

Try it out on a few sample problems!

Task 6

1. Prove that a tree with n vertices has $n - 1$ edges.

Base case: 1 vertex tree has 0 edges

Inductive Hypothesis: assume all trees with k vertices have $k - 1$ edges.

Inductive step: Consider a tree with $k + 1$ vertices. There must be some leaf, which has one edge. Remove the leaf and its edge. The resultant graph will still be acyclic, since removing an edge cannot create a cycle, and is connected because the edge we removed only connected the removed vertex to the rest of the graph. So, we now have a tree with k vertices, which by the IH has $k - 1$ edges. Put the removed vertex and edge back in, and we now have k edges. As our $k + 1$ tree was arbitrary, all trees with $k + 1$ vertices have k edges.

2. *Optional:* Prove that K_n has $\frac{n(n-1)}{2}$ edges. *Note that we could do this without build-down induction, but you should use it here for practice.*

Base case: K_1 has 0 edges, which is $\frac{1*0}{2}$

IH: Assume K_k has $\frac{k(k-1)}{2}$ edges

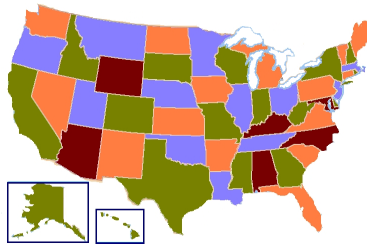
IS: Consider K_{k+1} . We can get K_k by removing a vertex and all its incident edges, of which there are k . By IH, K_k has $\frac{k(k-1)}{2}$ edges. If we add back in the vertex and edges, we get $\frac{k(k-1)}{2} + k = \frac{k(k-1)+2k}{2} = \frac{k(k-1+2)}{2} = \frac{k(k+1)}{2}$

edges as needed.

Optional: Graph Coloring

*Note: This entire section is optional, so feel free to get checked off now. Graph coloring will **not be on the final**, but we think it's a super interesting topic in graph theory!*

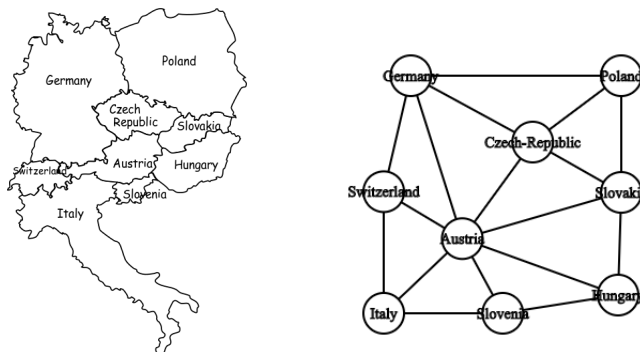
Maps are often drawn such that no two adjacent regions are the same color. This makes it easy when looking at the map to see the distinct regions. For instance, here is a map of the United States where no adjacent states are the same color:



You might notice that this map was drawn with only 4 distinct colors. There are 50 U.S. states, so it might be a bit surprising that this is possible. This leads to a question that cartographers and mathematicians have asked for centuries: given a map, how many colors will we need to draw it?

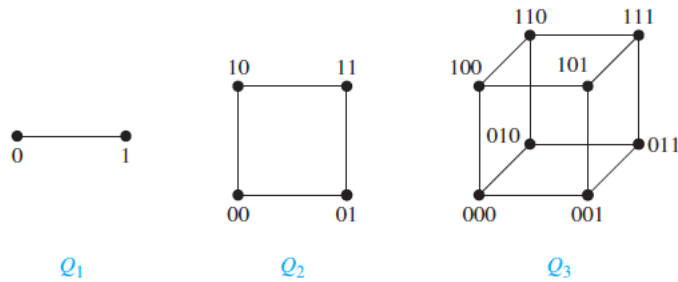
Like so much else in life, this is really a graph theory problem in disguise! Any map can be easily converted to a graph.

For instance, we can represent this map of a region of Europe with the displayed graph. The graph was created by making each country into a vertex, and drawing an edge between two vertices if the countries share a border.



Task 7

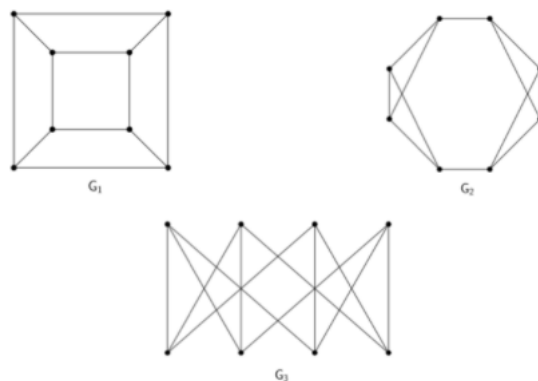
1. Recall the n -dimensional cube graph; in particular, consider Q_1 , Q_2 , and Q_3 , shown below:



What is the minimum number of colors we need to properly color Q_1 , Q_2 , and Q_3 ? What about for Q_n ?

We only need 2 colors for any Q_n ! For instance, we can color the vertices depending on if they have an even or odd number of '1's in their corresponding binary string, as labelled above.

2. In task 2, we determined which two out of these three graphs are isomorphic:



Graph coloring is often helpful in determining isomorphic graphs. Determine the minimum number of colors needed to properly color G_1 , G_2 , and G_3 .

G_1 and G_3 can be properly colored using only two colors, while G_2 needs a minimum of three colors. Thus, we know that G_2 cannot be isomorphic to G_1 or G_3 !

Checkoff: Call over a TA!